Answer Key — Immunity, Vaccines and Treatment

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1	**(b) bacteria** — antibiotics target bacteria, not viruses or protozoa.
Q2	**(b) preventive —** vaccines prevent disease; they do not cure an existing illness.
Q3	**(b) Edward Jenner —** Jenner used cowpox to protect against smallpox.
Q4	**(b) Alexander Fleming** — Fleming discovered penicillin from a mould in 1928.
Q5	**(a) Both true, R explains A** — avoiding misuse prevents resistance — exactly why we follow the prescription.
Q6	**immunity** — immunity is the body's defence ability.
Q7	**cowpox** — cowpox is the related, milder disease.

Short answer

Q8	A preparation of weakened/dead pathogen or its harmless part that trains the immune system to recognise and fight the germ, giving acquired immunity before infection.
Q9	The cold is caused by a virus; antibiotics act only on bacteria, so they cannot kill the virus.
Q10	When bacteria survive and multiply despite an antibiotic. Reduce it by using antibiotics only when prescribed, in the correct dose and full duration, and avoiding unnecessary use.

Long / application

Q11	(1) Antibiotics work only on bacteria, not viruses (cold, flu) or protozoa. (2) Overuse causes antibiotic resistance, making them useless. (3) Many diseases are non-communicable (diabetes, asthma) and need lifestyle changes, not antibiotics; prevention (hygiene, vaccines) still matters.
Q12	**Observation:** milkmaids who had cowpox didn't get smallpox. **Hypothesis:** something in cowpox protects against smallpox. **Experiment:** he gave cowpox material to a boy, then exposed him to smallpox. **Result:** the boy did not fall ill — the first vaccine, later helping eradicate smallpox.